



NORTH SYDNEY BOYS HIGH SCHOOL

2011 Year 11 Task 3

Mathematics Extension 1

General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Ms Collins
- ☐ Mr Fletcher/Ms Everingham
- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Weiss

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total %
Mark	$\overline{13}$	$\overline{10}$	$\overline{11}$	$\overline{11}$	$\overline{12}$	$\overline{57}$	$\overline{100}$

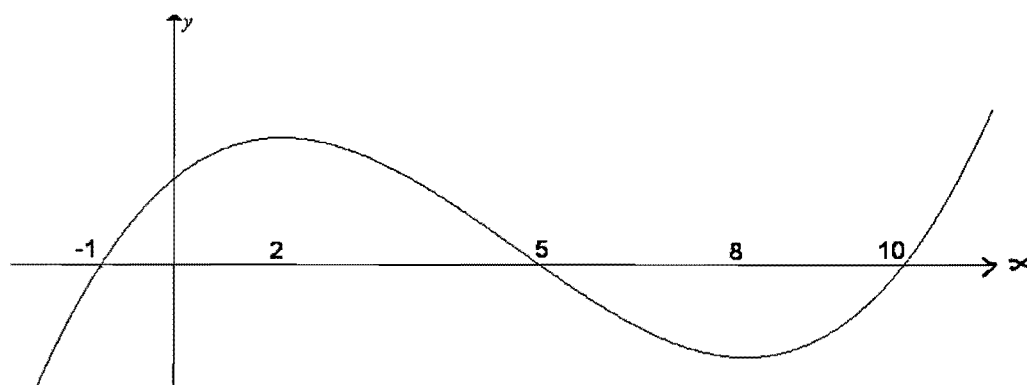
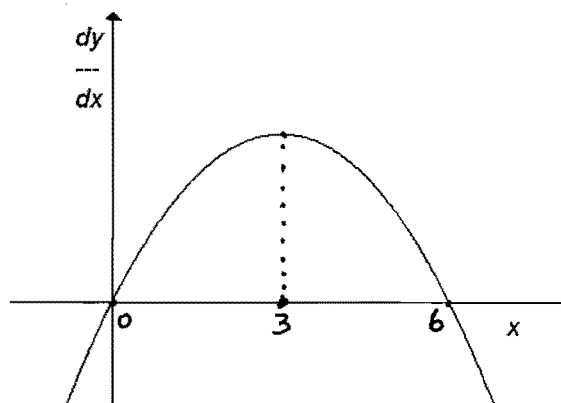
Question 1 (13 marks) Start a new page.**Marks**(a) Differentiate the following with respect to x :

(i) $y = x\sqrt{x}$ 2

(ii) $y = (2 - 3x^2)^4$ 2

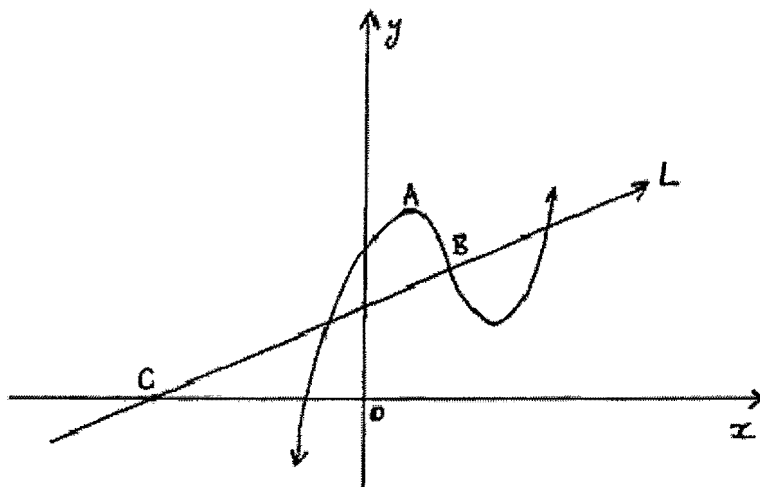
(iii) $y = 2x^5(x + 3)^9$ 2

(iv) $y = \frac{x^2}{x - 2}$ 2

(b) Differentiate from first principles the function $f(x) = x^2 + 5x$ 3(c) The graph of $y = f(x)$ is shown below. Draw a sketch of $y = f'(x)$. 2**Question 2 (10 marks)** Start a new page.(a) The curve $y = f(x)$ has gradient function $\frac{dy}{dx}$. The graph of $\frac{dy}{dx}$ as a function of x is shown:Find where the curve $y = f(x)$

- | | | |
|------|------------------------------|---|
| (i) | has a maximum turning point; | 1 |
| (ii) | is concave down. | 1 |

(b)



The diagram shows a sketch of $y = x^3 - 6x^2 + 9x + 4$. The curve has a local maximum at A and a point of inflexion at B . The line L is a normal to the curve at point B .

- | | |
|---|---|
| (i) Find the coordinates of point A and verify it is a maximum. | 3 |
| (ii) Find the coordinates of point B and verify it is an inflexion. | 2 |
| (iii) Find the equation of the normal (line L). | 3 |

Question 3 (11 marks) Start a new page.

(a) Evaluate:

- | | |
|---|---|
| (i) $\log_4\left(\frac{1}{\sqrt{2}}\right)$ | 2 |
| (ii) $\log_6 12$ (to 2 decimal places) | 2 |

(b) Using $\log_a 2 = 0.387$ and $\log_a 3 = 0.613$,

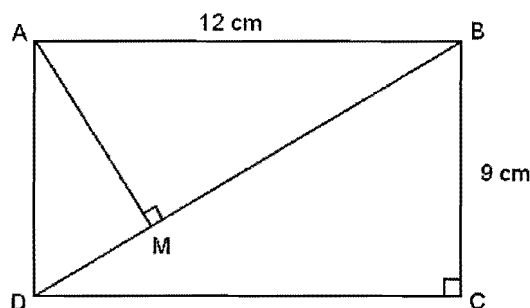
find the value of $\log_a 12$. 2

(c) Sketch $y = \log_2(2x - 2)$ 2

(d) Solve the equation $\log_2 x + \log_2(x + 7) = 3$ 3

Question 4 (11 marks) Start a new page.

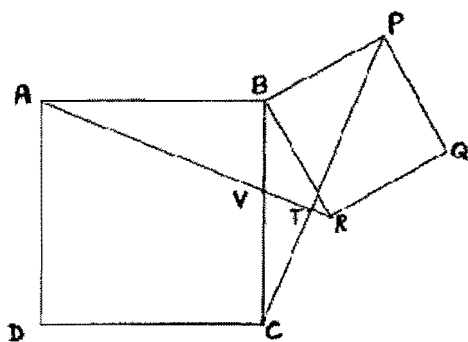
(a)



$ABCD$ is a rectangle. $AB = 12$ cm,
 $BC = 9$ cm, and AM is perpendicular to BD .
Copy the diagram to your answer sheet.

- | | |
|--|---|
| (i) Prove $\triangle ABM \sim \triangle DBA$ | 2 |
| (ii) Hence find the length of AM . | 3 |

- (b) In the diagram, $ABCD$ and $BPQR$ are squares. AR intersects PC at T , and BC at V .

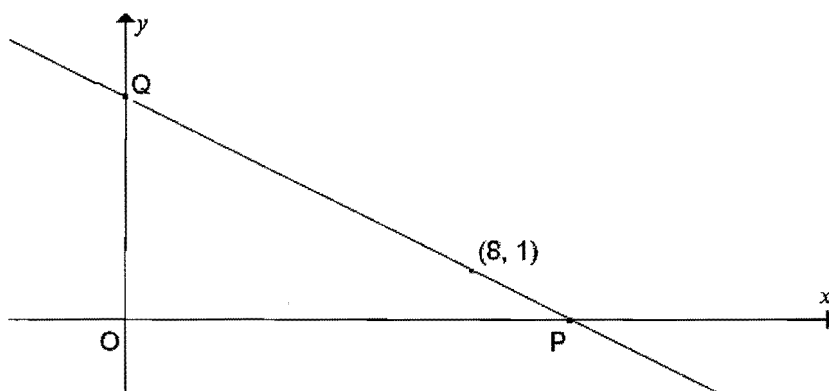


Copy the diagram to your answer sheet.

- (i) Prove that $\triangle ABR \equiv \triangle CBP$ 3
- (ii) Use this to prove $AR \perp PC$ 3

Question 5 (12 marks) Start a new page.

- (a) A line is drawn through the point $(8,1)$ to cut the positive x -axis at P and the positive y -axis at Q . The gradient of PQ is m .



- (i) Find the equation of PQ in terms of m . 1
- (ii) Show that the area of $\triangle OPQ$ is $A = \frac{1}{2}(16 - 64m - \frac{1}{m})$ 3
- (iii) Find the minimum area of $\triangle OPQ$ 4
- (b) A particle moves along the x -axis so that its displacement x , in metres, is given by $x = 4 + 6t - t^2$ where t is the time in seconds.
- (i) Express the particle's velocity, v , as a function of time, t 1
- (ii) When is the particle at rest? 1
- (iii) Find the total distance it moves in the first 7 seconds. 2

Q1 (a)

$$(i) \quad y = x\sqrt{x} \\ = x^{\frac{3}{2}}$$

$$\therefore \frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}}$$

✓ (e.g. uses indices)

✓ correct answer

$$(ii) \quad y = (2-3x^2)^4$$

$$\therefore \frac{dy}{dx} = 4 \cdot (2-3x^2)^3 \cdot -6x \\ = -24x (2-3x^2)^3$$

✓✓ uses chain rule correctly

$$(iii) \quad y = 2x^5(x+3)^9$$

$$\therefore \frac{dy}{dx} = (x+3)^9 \cdot 10x^4 + 2x^5 \cdot 9(x+3)^8 \\ = 10x^4(x+3)^9 + 18x^5(x+3)^8$$

✓✓ correct use of product rule

$$(iv) \quad y = \frac{x^2}{x-2}$$

$$\therefore \frac{dy}{dx} = \frac{(x-2) \cdot 2x - x^2 \cdot 1}{(x-2)^2} \\ = \frac{x^2 - 4x}{(x-2)^2}$$

✓✓ correct use of the quotient rule

$$(b) \quad f(x) = x^2 + 5x$$

$$\therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + 5(x+h) - [x^2 + 5x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2xh + h^2 + \cancel{5x} + 5h - \cancel{x^2} - \cancel{5x}}{h}$$

$$= \lim_{h \rightarrow 0} (2x + h + 5) = 2x + 5$$

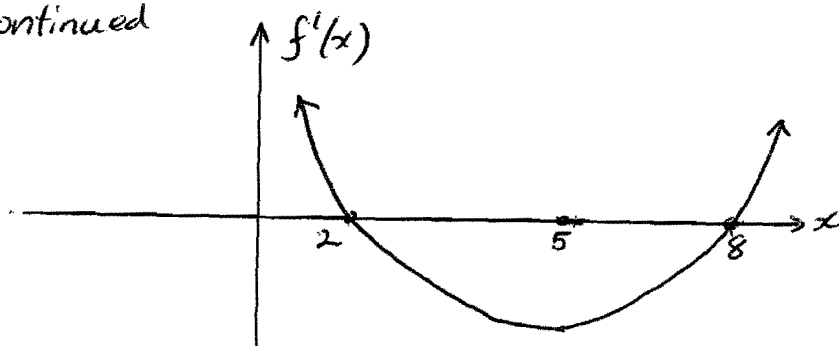
✓ correct start

✓ correct working

✓ correct answer

Q1-continued

(c)

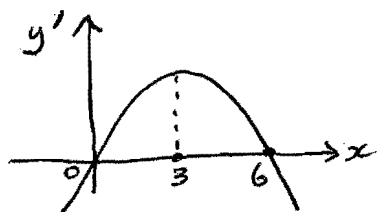


✓ correct shape
✓ both x-intercepts

13

Q2

(a)



(i) Turning points when $y' = 0$ $\therefore$ at $x=0, 6$

Test:

x	$- \epsilon$	0	$+ \epsilon$
y'	$-$	0	$+$
	$\backslash$	$-$	$/$

x	$6 - \epsilon$	6	$6 + \epsilon$
y'	$+$	0	$-$
	$/$	$-$	$\backslash$

$\therefore$ max. turning point at $x=6$

✓ for $x=6$

(ii) Concave down when $\frac{d^2y}{dx^2} < 0$

i.e. when $\frac{dy}{dx}$ is decreasing.

$\therefore$ concave down when $x > 3$

✓ for $x > 3$

(b) $y = x^3 - 6x^2 + 9x + 4$

(i) $y' = 3x^2 - 12x + 9$
 $= 3(x^2 - 4x + 3)$
 $= 3(x-3)(x-1)$

$y'' = 6x - 12$
 $= 6(x-2)$

✓ correct derivative

$\therefore$ Turning points when $y' = 0$ $\therefore$ at $x=1, 3$.

$\begin{cases} y''(1) = 6(1-2) < 0 \\ y''(3) > 0 \end{cases}$ $\therefore$ max. turning point at $x=1$.

Thus $A = (1, 8)$

✓ testing performed

✓ coordinates of A

Q2 - continued(b) (ii) Possible inflexions when $y'' = 0$

$$\therefore 6(x-2) = 0 \quad \therefore x = 2$$

Test:

x	1.9	2	2.1
y''	-	0	+

Changes concavity $\therefore$ inflexion at $x=2$.

$$\text{Thus } B = (2, 6)$$

$$(iii) \quad y' = 3(x-3)(x-1)$$

$$\therefore y'(2) = 3(-1)(1) = -3$$

$$\therefore m_T \text{ at } B \text{ is } -3$$

$$\therefore m_N \text{ at } B \text{ is } \frac{1}{3}$$

$$\text{Thus normal is } y - 6 = \frac{1}{3}(x - 2)$$

$$\text{i.e. } 3y - 18 = x - 2$$

$$\therefore x - 3y + 16 = 0$$

$$(\text{or } y = \frac{1}{3}x + \frac{16}{3})$$

✓ correct inflexion point (x-value)

✓ Tests for change in concavity.

✓ correct m_N

✓✓ Correct equation (using correct B coordinates)

10

Q3 (a)

$$\begin{aligned}
 (i) \quad \log_4\left(\frac{1}{\sqrt{2}}\right) &= \log_4\left(2^{-\frac{1}{2}}\right) \\
 &= \log_4\left([4^{\frac{1}{2}}]^{-\frac{1}{2}}\right) \\
 &= \log_4\left(4^{-\frac{1}{4}}\right) \\
 &= -\frac{1}{4}
 \end{aligned}$$

$$(ii) \quad \log_6 12 = \frac{\log_{10} 12}{\log_{10} 6}$$

$$\doteq 1.39 \quad (2 \text{ d.p.})$$

✓ correct start

✓ correct answer

✓ Change of base formula

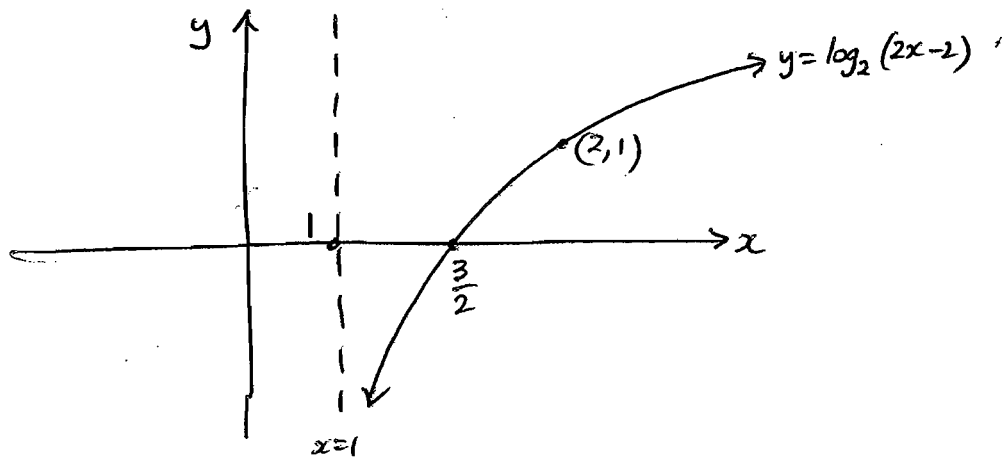
✓ correct answer (don't penalise d.p.'s)

Q3 - continued.

$$\begin{aligned}
 (b) \quad \log_a 12 &= \log_a (2^2 \cdot 3) \\
 &= 2 \log_a 2 + \log_a 3 \\
 &= 1.387
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad y &= \log_2 (2x-2) \\
 &= \log_2 2[x-1]
 \end{aligned}$$

[Note: curve is $\log_2 x$ shifted to right & raised up.]
 Thus vertical asymptote at $x=1$.
 Also, curve = 0 when $2x-2=1 \therefore x = \frac{3}{2}$ is x-int.]



✓ correct shape & asymptote

✓ correct x-int. or other point on curve.

$$(d) \quad \log_2 x + \log_2 (x+7) = 3$$

$$\therefore \log_2 x(x+7) = 3$$

$$\therefore x(x+7) = 2^3 = 8$$

$$\therefore x^2 + 7x - 8 = 0$$

$$(x+8)(x-1) = 0$$

$$x = -8 \text{ or } x = 1$$

But $x > 0$, $\therefore$ solution is $x = 1$

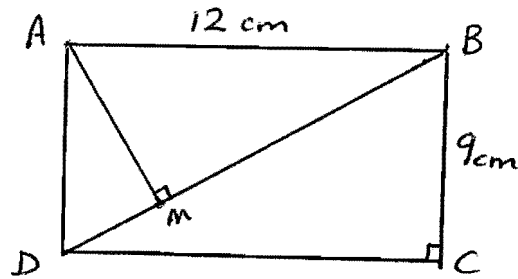
✓ uses log laws correctly

✓ gets -8, 1

✓ correct solution

Q4

(a)

(i) In $\triangle ABM$, $\triangle DBA$,

$$\angle ABM = \angle DBA \text{ (common angle)}$$

$$\angle AMB = 90^\circ = \angle DAB \text{ (angle of rectangle)}$$

$$\therefore \triangle ABM \parallel \triangle DBA \text{ (equiangular)}$$

$$(iii) \quad \frac{AM}{AD} = \frac{AB}{DB} \text{ (corresp. sides in proportion in similar } \triangle's)$$

$$\therefore AM = \frac{AD \cdot AB}{DB} = \frac{(9)(12)}{DB}$$

$$\text{But } DB = \sqrt{9^2 + 12^2} = 15$$

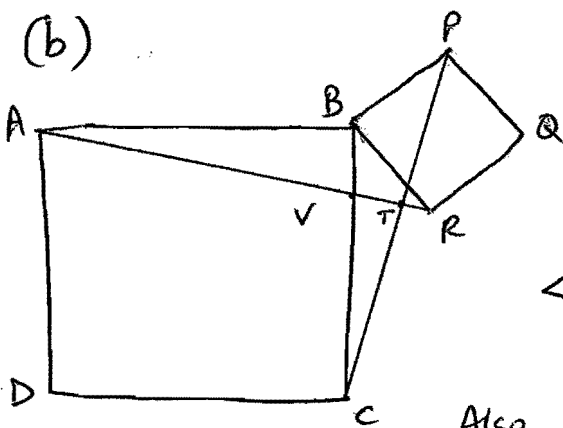
$$\therefore AM = \frac{108}{15} = 7.2$$

✓ correct reasoning
✓ correct similarity condition
[* no abbrev.]

✓ correct proportion statement / equation
✓ correct DB

✓ correct answer

(b)

(i) In $\triangle ABR$, $\triangle CBP$,

$$AB = CB \text{ (sides of square } ABCD)$$

$$BR = BP \text{ (sides of square } BPRQ)$$

$$\begin{aligned} \angle ABR &= \angle ABC + \angle CBR \\ &= 90^\circ + \angle CBR \text{ (} \angle \text{ of square)} \end{aligned}$$

$$\begin{aligned} \text{Also, } \angle CBP &= \angle RBP + \angle CBR \\ &= 90^\circ + \angle CBR \\ &= \angle ABR \end{aligned}$$

$$\therefore \triangle ABR \equiv \triangle CBP \text{ (SAS)}$$

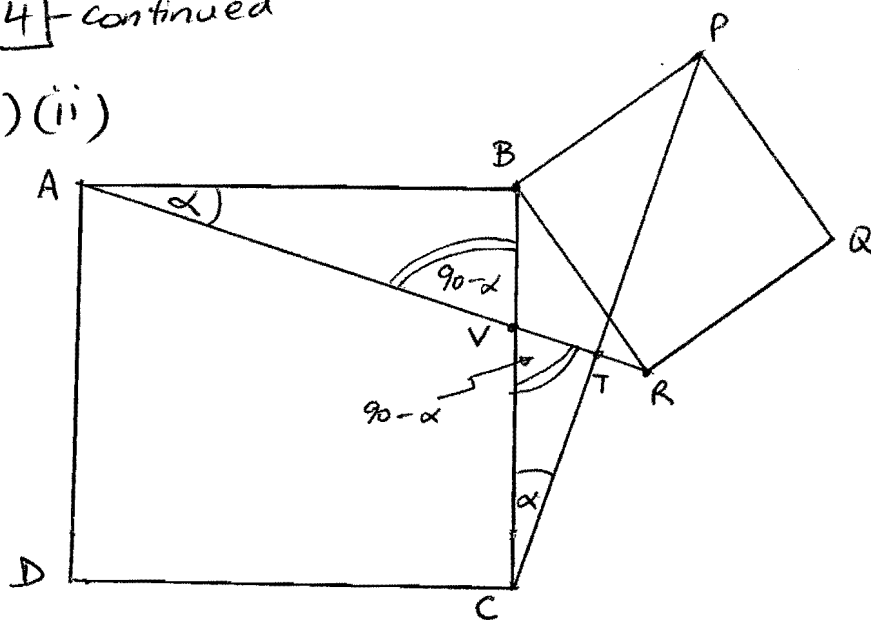
✓ correct sides info.

✓ correct angle info.

✓ correct (* reasons are needed for full marks)

Q4 - continued

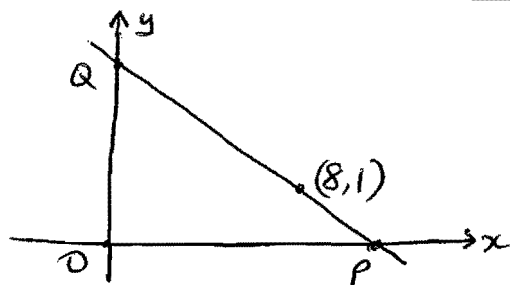
(b)(ii)

Let $\angle BAV = \alpha$.Then $\angle BCP = \alpha$ [corresp. angles in
congruent Δ 's]Then $\angle BVA = 90^\circ - \alpha$ [angle sum in ΔABV] $\therefore \angle CTV = 90^\circ - \alpha$ [vert. opposite $\angle$'s are equal]Thus $\angle VTC = 180^\circ - \alpha - (90^\circ - \alpha)$ [angle sum
in ΔVTC]
 $= 90^\circ$ i.e. $AR \perp PC$ ✓ uses the
congruence
correctly
[mandatory]✓✓ correct
argument
with
reasons

11

Q5

(a)



$$(i) \quad y - 1 = m(x - 8)$$

✓ correct answer

$$(ii) \quad \text{At } Q, x = 0 \therefore y - 1 = m(-8) \\ y = -8m + 1$$

✓ correct Distance OQ

$$\text{At } P, y = 0 \therefore -1 = mx - 8m \\ x = 8 - \frac{1}{m}$$

✓ correct Distance OP

$$\therefore \text{Area } \triangle OPQ = \frac{1}{2} \cdot (1 - 8m) \left(8 - \frac{1}{m}\right) \\ = \frac{1}{2} \left(8 - \frac{1}{m} - 64m + 8\right) \\ \therefore A = \frac{1}{2} \left(16 - 64m - \frac{1}{m}\right)$$

✓ Derives A correctly

$$(iii) \quad \frac{dA}{dm} = \frac{1}{2} \left[-64 + \frac{1}{m^2}\right]$$

✓ correct $\frac{dA}{dm}$

$$\text{For minimum, } \frac{dA}{dm} = 0 \therefore \frac{1}{m^2} = 64$$

$$\therefore m = \pm \frac{1}{8}$$

$$\text{But } m < 0 \therefore m = -\frac{1}{8}$$

✓ for $m = -\frac{1}{8}$

$$\text{Test: } \frac{d^2A}{dm^2} = \frac{1}{2} \left[\frac{-2}{m^3}\right] = \frac{-1}{m^3} \\ \therefore = -\frac{1}{\left(-\frac{1}{8}\right)^3} > 0$$

✓ Tests it's a minimum

$\therefore m = -\frac{1}{8}$ represents a minimum area.

The minimum area is thus

$$A = \frac{1}{2} \left[16 - 64\left(-\frac{1}{8}\right) - \frac{1}{-\frac{1}{8}}\right] \\ = \frac{1}{2} [16 + 8 + 8] = 16 \text{ units}^2.$$

✓ correct minimum area

Q5 - continued

(b) $x = 4 + 6t - t^2$

(i) $v = \frac{dx}{dt} = 6 - 2t$

(ii) "at rest" $\Rightarrow v = 0$

$\therefore 6 - 2t = 0$

$t = 3 \text{ seconds.}$

(iii) At $t = 0$, $x = 4$

At $t = 3$, $x = 13$

At $t = 7$, $x = -3$

The particle starts at $x = 4$, moves right until it comes to rest momentarily at $x = 13$, then moves left to reach $x = -3$ after 7 seconds.

$$\therefore \text{Total distance} = (13 - 4) + 13 + 3$$

$$= 25 \text{ metres}$$

✓ correct working

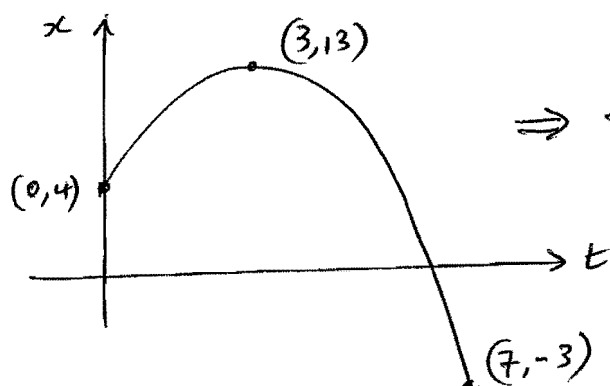
✓ correct answer.

Alternatively,

$x = -(t^2 - 6t - 4)$

$= -(t^2 - 6t + 9 - 13)$

$\therefore x = -(t - 3)^2 + 13$



$$\Rightarrow \text{Total distance} = 9 + 16 = 25 \text{ m.}$$